Major Research and Education Activities

Our goal is to implement a wireless-networked, distributed heterogeneous environment on campus and use it to conduct research in databases, networking, distributed systems, and multimedia. Our focus is on large-scale systems in which data is the critical resource and system services are based on various data manipulation functions including data collection, movement/delivery, aggregation/processing, and presentation. Specific research issues that we have investigated include content-based access, personalized views, multi-dimensional indexing, smart end-to-end applications, joint source-network coding, scalable storage, reliable network service, information summarization, distributed collaboration, multimedia annotation, and interactivity.

Our progress in deploying a wireless network has been significant. We now have high quality wireless coverage in all of the faculty offices and research laboratories associated with this grant. In addition, we have wireless access in the digital classroom and other teaching classrooms around the campus. We have our primary research network, as well as a secondary network for use with courses incorporating wireless networking assignments.

The Digital Classroom is operational and has been used for a number of seminars and courses. Significant changes have been made to enhance audio and video capabilities in the classroom: digital mixer with echo cancellation, real time editing of video, installation of an access grid node, creation of secondary classroom, content distribution in the classroom, and building of a classroom services architecture.

We have recruited a staff to oversee the project. We have also built an infrastructure for supporting a *lecture on demand/* application. As a part of this infrastructure, we have designed feature extraction techniques and new index structures.

The main research is being pursued in five main areas and is discussed later.

A. Multimedia Information Processing. This research is investigating topics such as creation of structured documents from handwritten lecture notes, relevance feedback, perceptual distance functions, and virtual I/O.

B. Networks. This research is investigating topics such as hybrid networks and QoS for wireless data.

C. Databases & Distributed Systems. This research is investigating topics such as load balancing, data replication, mobile middleware, efficient search and retrieval, cluster computing, and information summarization.

D. Digital Classroom. This part of the project is developing new infrastructure for teaching and education, and examining the appropriate architecture for distributed and distance learning.

E. Collaborative Computing. This research is investigating topics such as visualization tools, and sticky notes.